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# Francisco Ríos Fierro

**Nationality:** Chilean **Date of birth:** 30/08/1999 **Gender:** Male

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## ABOUT ME

Data expert and full-stack developer at the Centre for European Policy Studies in Brussels, working in data analysis, data visualisation and open data. I am the principal architect of AI World (aiworld.eu), CEPS's global AI observatory, backend and frontend. I have advanced, hands-on proficiency in JavaScript and JSON for creating highly tailored and dynamic web behaviours and custom visual configurations. I render complex, accessible and interactive statistical charts for the web. I parse and manipulate MS Excel, CSV and JSON files for data ingestion and pipeline integration. My work has been presented to the European Commission and has fed projects funded by the EC AI Office, the United Nations and the Microsoft AI Economy Institute. I hold an MSc in Industrial Engineering, with research in optimisation and machine learning.

## WORK EXPERIENCE

***Centre for European Policy Studies*** – Brussels, Belgium

### Data scientist

[ 01/01/2026 – Current ]

- Principal architect of AI World, backend and frontend, built with Next.js, React and PostgreSQL.
- Advanced, hands-on JavaScript and JSON to create highly tailored, dynamic web behaviours and custom visual configurations for the AI World dashboards.
- Rendering of complex, accessible and interactive statistical charts in D3.js, including charts for NVIDIA's GTC events. Analysis I helped develop was cited three times by Jensen Huang in GTC keynotes.
- Parsing and manipulation of MS Excel, CSV and JSON files for data ingestion and pipeline integration, in the OpenAlex research pipeline (Python) and the WIPO PCT patents pipeline (FastAPI and SQLite).
- Data analysis with GitHub researchers on agentic AI and with Hugging Face researchers on open-source AI and AI accelerators.
- Built the daily AI model leaderboard on the AI World homepage.
- Designed the AI World chart house styles and user interfaces, with accessible markup (ARIA).

***Centre for European Policy Studies*** – Brussels, Belgium

### Junior Data Scientist

[ 01/03/2025 – 01/01/2026 ]

- Built new features for AI World and improved its layout on phones.
- Built Python pipelines that collect Hugging Face model, dataset and paper data.
- Contributed to the State of Open-Source AI 2025 work with Hugging Face.
- Ran large LLM batch jobs with the OpenAI and Anthropic APIs to classify data.
- Built multiple crosswalks between classification systems.

***Centre for European Policy Studies*** – Brussels, Belgium

### Intern Data Scientist

[ 01/09/2024 – 01/03/2025 ]

- Contributed to the State of Open-Source AI 2024 work with Hugging Face.
- Built multiple interactive D3.js charts and hosted them in AI World.
- Cleaned and matched datasets.

***Department of Industrial Engineering, Faculty of Engineering, University of Concepción*** – Concepción, Chile

### Project Supporting Engineer

[ 01/05/2024 – 31/08/2024 ]

- Supported departmental projects for the Industrial Engineering programme.
- Automated key processes in Python. Examples are matching faculty advisors to graduation projects and enrolment for courses with limited places.
- Contributed to a new curriculum aligned with industry practice.

***Innovation Unit, Faculty of Engineering, University of Concepción*** – Concepción, Chile

### Data Scientist

[ 01/10/2023 – 31/08/2024 ]

- Built and ran a machine learning model that identifies first- to third-year engineering students at risk of dropping out.
- The faculty adopted the model as its standard early-warning instrument.

***Universidad de Concepción*** – Concepción, Chile

### Teaching assistant

[ 2020 – 2024 ]

- Teaching assistant for the Programming, Simulation and Machine Learning courses.

## EDUCATION AND TRAINING

### Industrial Engineering

***University of Concepción*** [ 01/03/2018 – 31/08/2023 ]

**City:** Concepción | **Country:** Chile | **Website:** www.udec.cl | **Level in EQF:** EQF level 6

### MSc in Industrial Engineering

***Universidad de Concepción*** [ 01/03/2023 – 31/05/2025 ]

**City:** Concepción | **Country:** Chile | **Website:** www.udec.cl | **Level in EQF:** EQF level 7

## LANGUAGE SKILLS

**Mother tongue(s):** Spanish

**Other language(s):**

### English

**LISTENING** C2 **READING** C2 **WRITING** C2

**SPOKEN PRODUCTION** C2 **SPOKEN INTERACTION** C2

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user*

## SKILLS

**Web development**

JavaScript / TypeScript / React / Next.js / HTML and CSS

**Data visualisation**

D3.js / d3plus / Interactive dashboards for the web

**Data ingestion and pipelines**

JSON parsing and manipulation / CSV parsing and manipulation / Python / pandas / SQL / SQLite / Prisma / MS Excel parsing and manipulation (openpyxl)

**Backend and infrastructure**

REST APIs / Docker / Nginx / Git and GitHub

**AI and machine learning**

LLM classification pipelines / Text embeddings

## PROJECTS

[ 03/2025 – Current ]
**AI World, global AI observatory** Principal architect of the platform. Supported by Google.org.

**Link:** https://aiworld.eu

[ 10/2026 – Current ]
**Past And Current Transitions (PACT): A new pact for fair and resilient European regions** Horizon Europe Research and Innovation Action, 36 months. I am helping with the design of its visual identity.

[ 01/02/2026 – Current ]
**Artificial Intelligence Trends Across NVIDIA Core Domains** I built and hosted a weekly LLM classification pipeline over OpenAlex to offer insights to NVIDIA researchers.

**Link:** https://aiworld.eu/agentic-map/

[ 15/12/2025 – Current ]
**BEL-AI - Leveraging AI for Belgium** Ran the data pipelines necessary for the project, in addition to designing and building the visualisations.

**Link:** https://www.ceps.eu/ceps-projects/bel-ai/

## PUBLICATIONS

[2025]
**Surgery scheduling problem considering the affinity and preferences in the surgical team**

**Journal Name**: Health Care Management Science

Ríos-Fierro, F., Latorre-Núñez, G., & Contreras-Bolton, C. (2025). Surgery scheduling problem considering the affinity and preferences in the surgical team. Health Care Management Science, 28(4), 842-865.

## HONOURS AND AWARDS

[ 14/06/2026 ]

**First prize, Multi-Agent & Autonomous Coding AI hackathon co-organised by NVIDIA (Amsterdam)**

[ 28/03/2026 ]

**First prize, OpenClaw AI hackathon co-hosted by Anthropic (Amsterdam)**